



POLITECNICO
MILANO 1863

ASSIGNMENT 1

STRUCTURAL DYNAMICS AND AEROELASTICITY

Images and Comments

POLITECNICO DI MILANO

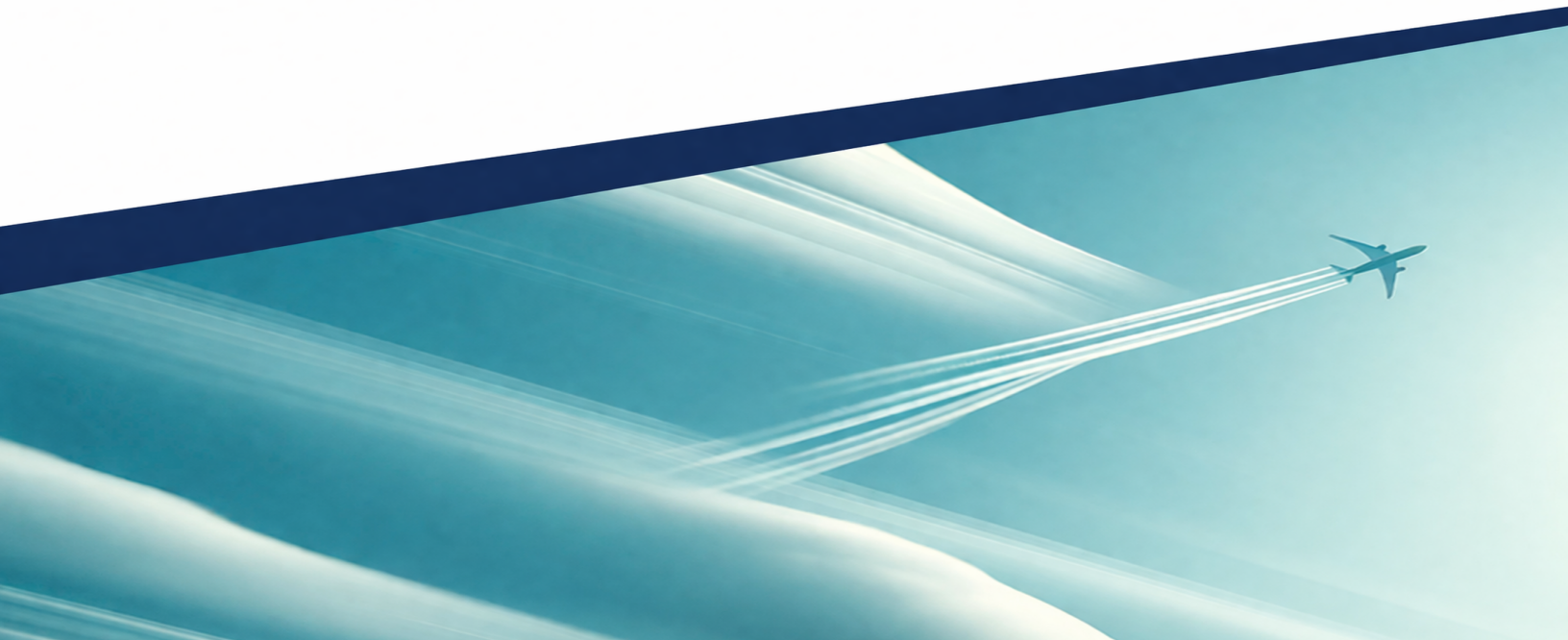
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Influence of strut position on divergence

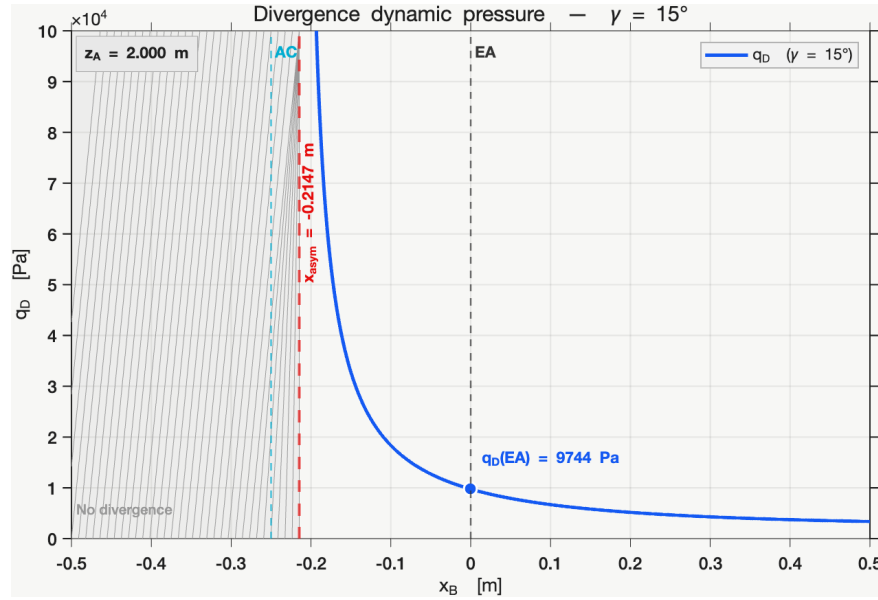


FIGURE 1 – Divergence dynamic pressure vs. strut position x_B for $\gamma = 15^\circ$.

Figure 1 shows that q_D is highly sensitive to x_B . A vertical asymptote appears at $x_B \simeq -0.2147$ m. To its right, divergence occurs at positive dynamic pressure and q_D drops rapidly as x_B approaches the elastic axis, remaining finite there ($q_D \approx 9.7 \times 10^3$ Pa). To its left, no positive q_D exists : the strut introduces sufficient flexural–torsional stabilization to suppress static divergence entirely.

Limits of the no-divergence region. This region suppresses static divergence only within the present linear model and provides no guarantee against flutter or control reversal. Structurally, such placement may increase strut loads, impose mass penalties, or introduce buckling risk. Sensitivity to geometric tolerances is also a concern near the asymptotic boundary. A more complete nonlinear aeroelastic analysis is required to validate these conclusions.